QRFL Complete Technical Report

Threat-Timeline-Driven Quantum-Resistant Federated Learning
for Blockchain-Enabled Healthcare Systems

Step-by-step architecture · Pipeline · Measured results · Graphs · Analysis · Performance outcomes

Item	Value
Corpus date	2026-09-03
FL scale	10 seeds × 50 rounds × 270 configs
PQC backend	quantcrypt-cffi (PQClean)
Fabric	HLF 2.5 — 3 peers + orderer + CouchDB
Artifacts	qrfl-artifacts/ · docs/

1. Executive Summary

QRFL integrates (1) quantum-threat timeline forecasting, (2) NIST PQC benchmarking, (3) healthcare federated learning under Classical / Hybrid / Native PQ modes, (4) Hyperledger Fabric validation, and (5) Mosca/MOSCoW migration planning. The scientific contribution is the end-to-end integration, not any single component alone.

Primary outcomes. Predictive metrics are identical across security modes (TOST equivalent). FL round-latency differences are not significant after Holm correction. Native PQ HLF simulation overhead is ~4% (80.8 → 84.2 ms). Bootstrap 95% lower bound for ECDSA-256 threshold crossing is ~2042.1 under baseline η.

Metric	Result
IID-25 accuracy (all modes)	65.3%
IID-25 AUROC	0.566
Native PQ FL round overhead	Typically < 1% (NS after Holm)
HLF native total latency	84.2 ms (vs 80.8 classical)
Forecast planning trigger	~2042.1 (bootstrap 95% lower)
FL determinism check	passed

2. Five-Layer Process Architecture

The framework is organized as a layered stack. Lower layers produce quantitative constraints (crossing years, primitive latencies) that parameterize upper layers (FL modes, Fabric simulation, migration roadmap).

Layer	Name	Process role	Primary outputs
L1	Quantum-Threat Forecasting	When to migrate	forecast_validation.tex, scenarios
L2	PQC / Crypto-Agility	What algorithms	crypto_benchmarks.tex, ablation
L3	Healthcare FL	Utility & FL cost	fl_*.tex, all_results.csv
L4	Blockchain Validation	Ledger cost	ledger_latency.tex, docker_stats
L5	Migration Decision Support	Roadmap action	Mosca + MOSCoW prose

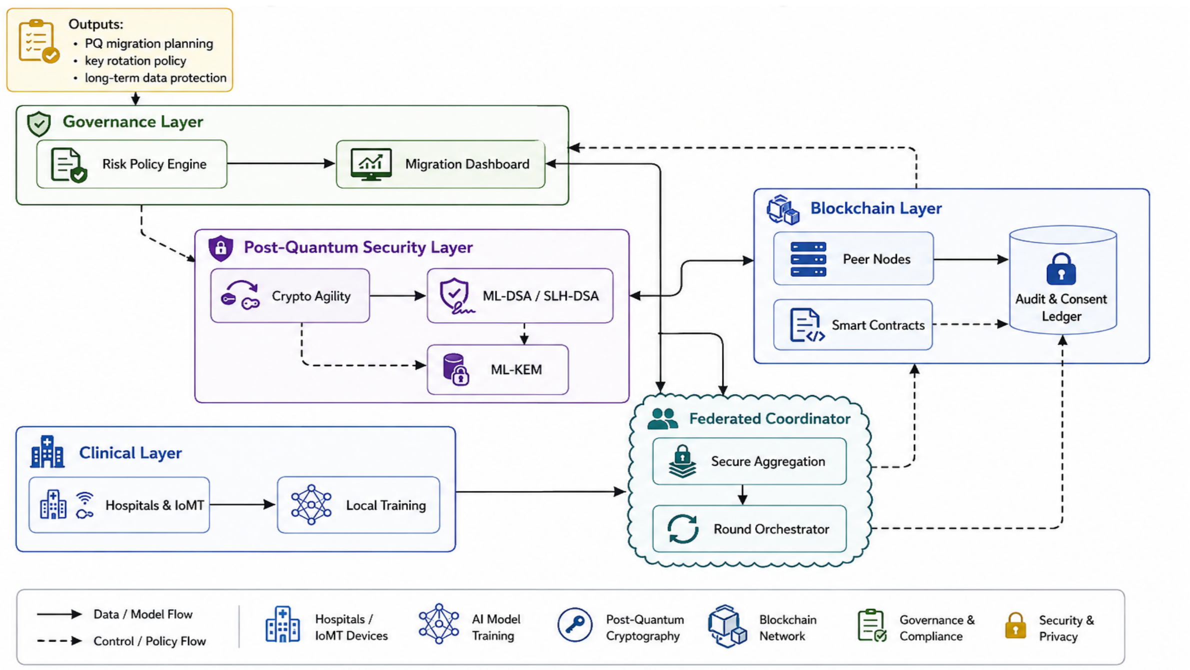

Figure: Layered QRFL system architecture (manuscript Fig. 2).

3. Step-by-Step Reproducibility Pipeline

Orchestrator: `run_all.ps1` / `run_all.sh`. Each stage merges macros into `results/results_macros.tex`.

Step	Module	Produces
1	forecasting.run_all	CSV + forecast_validation.tex + macros
2	pqc_benchmarks.run_benchmarks	trials/summary + crypto + ablation .tex
3	federated_learning.run_experiment	all_results.csv + fl_*.tex (hours)
4	resource_profiling.run_profile	resource_profiling.tex (energy optional)
5	statistical_tests.run_all	statistical_tests.tex (needs FL)
6	figures.generate_all	figures/output PDF/PNG
7	capture_hardware.py	hardware_specs.json

Optional Fabric (outside 1–7): generate_crypto → docker compose up → submit_transactions (calibrated simulation) → collect_metrics.

4. Protocol Flows (Runtime Architecture)

Flow	Mechanism	Analysis
F1 Hybrid certs	ECDSA + ML-DSA in X.509	~6.2× cert size; cache/fragment on IoMT
F2 Hybrid KEM	ML-KEM-768 ■ ECDHE → HKDF	HNDL-safe if ML-KEM holds
F3 Masking	Pairwise modular masks Σ=0	Utility invariant across modes
F4 Sign+endorse	ML-DSA-65 + Fabric 2-of-3	Sim total +~4% latency
F5 Revocation	DCRL + policy algo swap	Architectural; not timed

5. Federated Learning — Step-by-Step Round

Step	Function	Effect
1 Data	load_pneumoniamnist	Train/val/test loaders
2 Partition	dirichlet_partition	Client shards by α
3 Local train	train_local × clients	5 epochs Adam / round
4 Attack	apply_attack (optional)	Label-/sign-flip
5 Mask	apply_mask	Pairwise seeds; Σ masks = 0
6 Aggregate	fedavg | median | krum	Global CNN weights
7 Crypto tax	crypto_round_overhead	KEM+sig × N (from Exp A)
8 Evaluate	evaluate	Acc, AUROC, F1, latency

Experiment grid (270 configs): B-2 Dirichlet α×modes @25 clients; B client scaling {5,10,50}×modes; D Median/Krum and Byzantine attacks under native PQ. Full scale: 10 seeds × 50 rounds.

6. Results — Federated Learning Performance

6.1 Latency by security mode

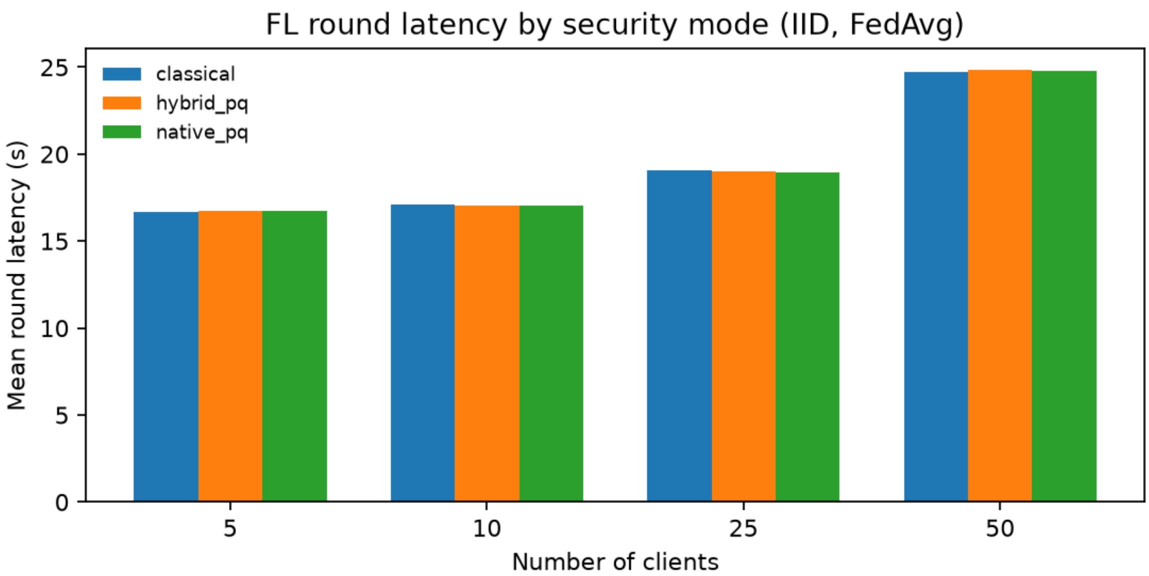

Figure: Mean FL round latency (IID FedAvg) by client count and mode.

Clients	Classical (s)	Hybrid (s)	Native (s)	Native overhead
5	16.70 ± 0.18	16.73 ± 0.18	16.73 ± 0.18	+0.17%

Clients	Classical (s)	Hybrid (s)	Native (s)	Native overhead
10	17.09 ± 0.17	17.07 ± 0.20	17.06 ± 0.15	–0.15%
25	19.06 ± 0.28	19.01 ± 0.29	18.62 ± 0.32	–2.30%
50	24.70 ± 0.37	24.82 ± 0.24	24.78 ± 0.37	+0.34%

6.2 Accuracy vs scale

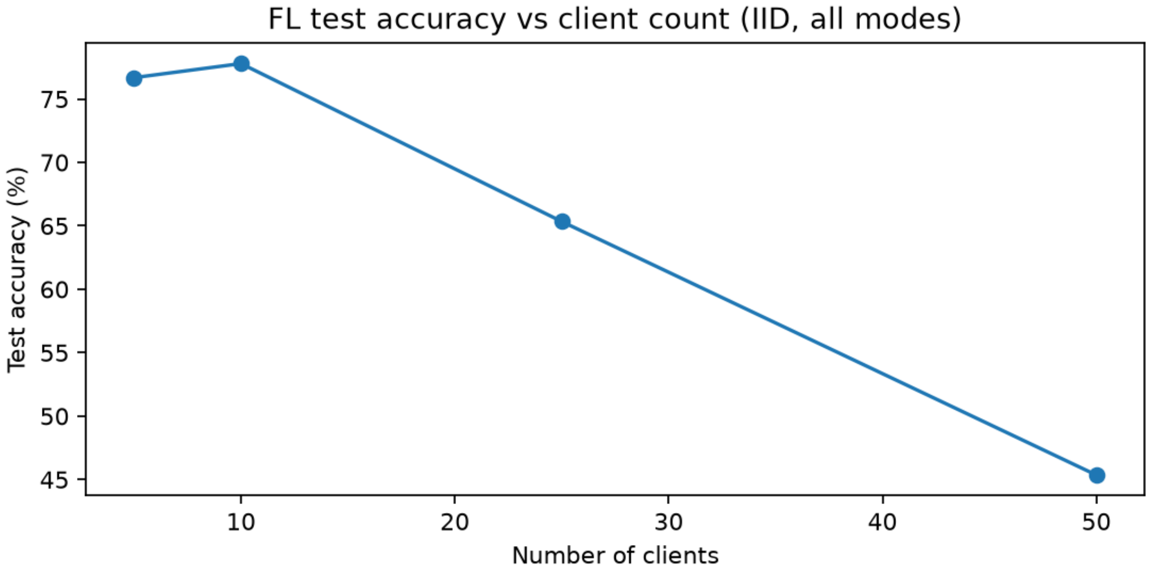

Figure: Test accuracy vs client count (IID; identical across modes).

Analysis. Accuracy peaks at 10 clients (77.8%) and falls at 50 clients (45.4%) due to data fragmentation—not PQC. At default 25 clients, all modes achieve 65.3% accuracy (determinism check passed).

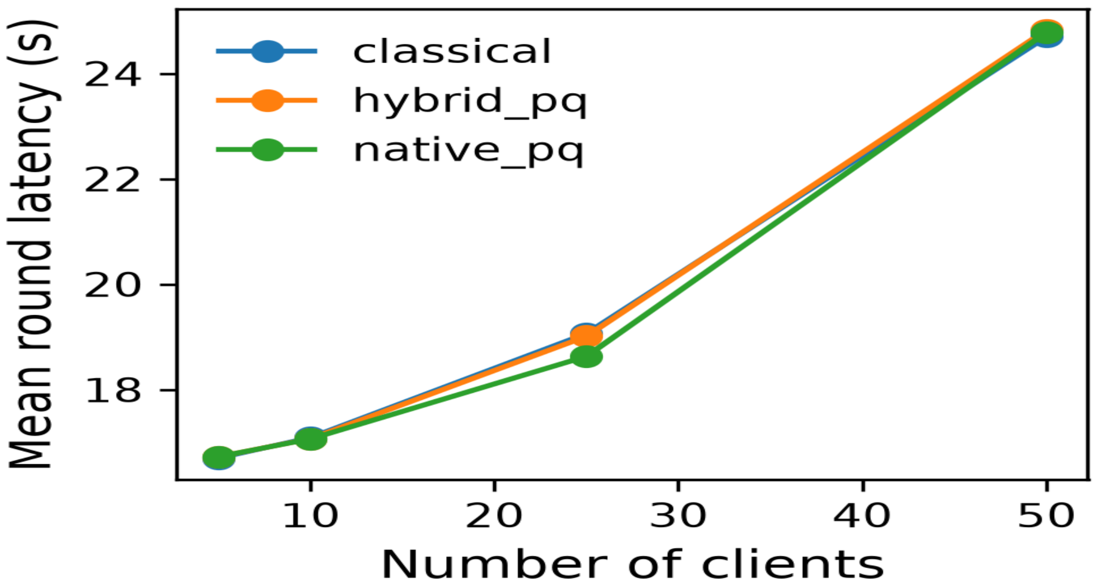

Figure: Pipeline-generated FL overhead chart (figures/output).

6.3 Non-IID and Byzantine outcomes

α	Accuracy	F1	Latency (s)
0.1	65.30%	0.767	19.18

α	Accuracy	F1	Latency (s)
0.5	65.42%	0.768	19.12
1.0 (IID)	65.34%	0.767	19.01

Attack	Mal %	Agg	Accuracy	Latency (s)
None	0%	FedAvg	65.98%	19.31
None	0%	Median	37.50%	18.39
Label flip	10%	FedAvg	37.45%	19.01
Label flip	20%	FedAvg	41.15%	19.01
Sign flip	20%	Krum	37.50%	18.52

Analysis. Under 50 rounds, non-IID α has limited accuracy impact. Byzantine label-flip severely degrades FedAvg; Median/Krum remain near 37.5% (often degenerate on this imbalanced task)—robust aggregation needs further tuning. PQ overhead remains secondary.

7. Results — PQC Micro-benchmarks (Exp A)

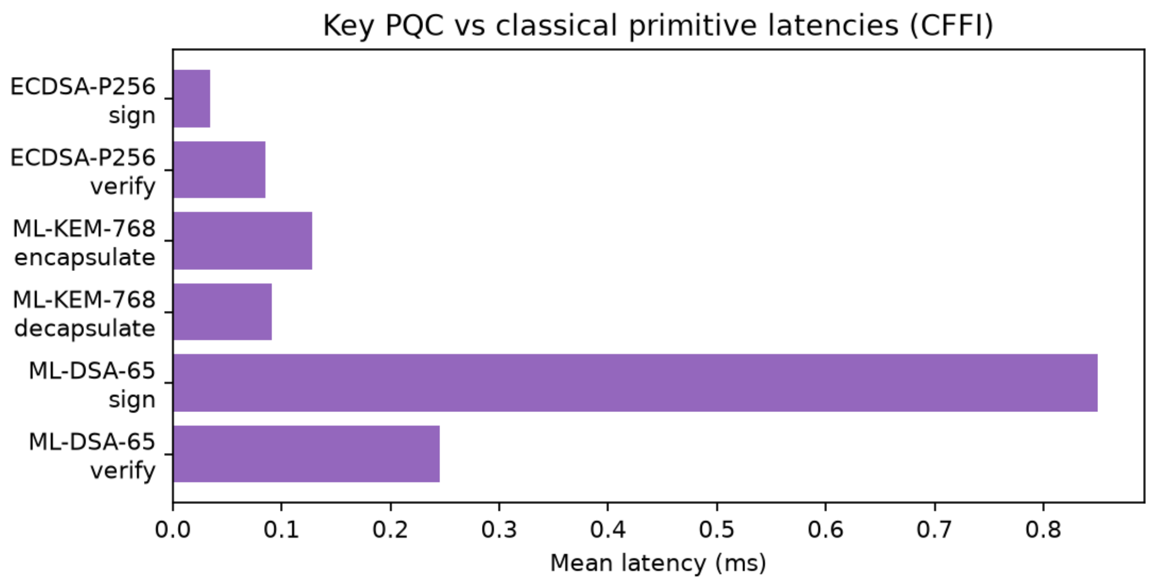

Figure: Key classical vs PQC operation latencies (CFFI, 10k trials).

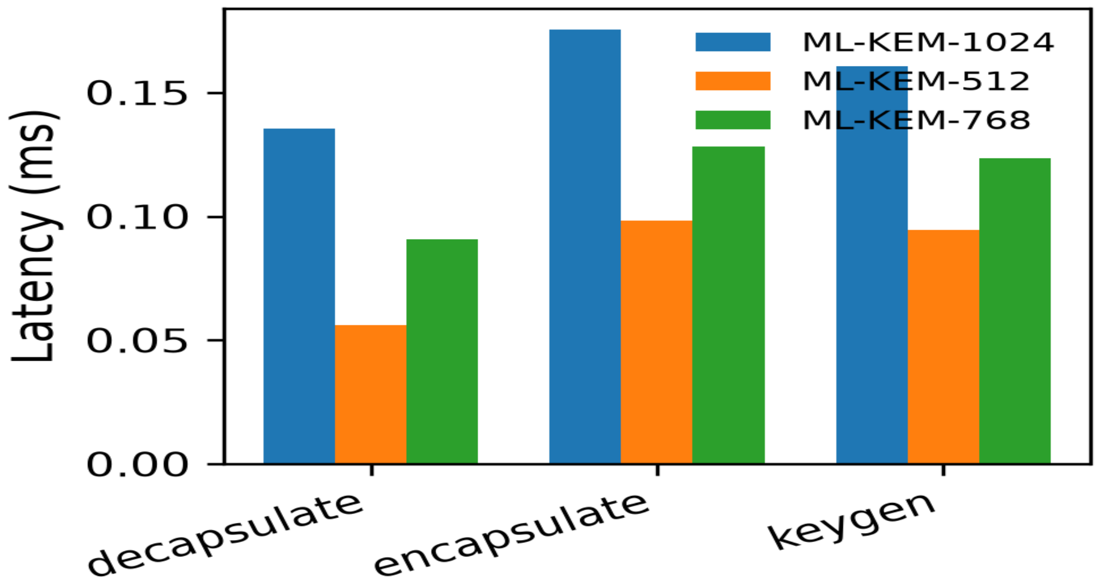

Figure: ML-KEM encaps/decaps overhead chart.

Scheme	Op	Mean (ms)	SD (ms)
ECDSA-P256	sign	0.035	0.002
ECDSA-P256	verify	0.085	0.003
ML-KEM-768	encaps	0.128	0.004
ML-KEM-768	decaps	0.091	0.013
ML-DSA-65	sign	0.850	0.502
ML-DSA-65	verify	0.245	0.019
SLH-DSA-256s	sign	536.8	5.84

Analysis / design choice. Default profile is ML-KEM-768 / ML-DSA-65 (NIST Level 3). SLH-DSA is reserved for long-term attestation due to multi-hundred-ms signing cost. Exp F (quantcrypt API) shows higher wrapper latencies (~3–5 ms) and ~73–76 MB RSS; energy unavailable without RAPL.

8. Results — Blockchain (Exp C)

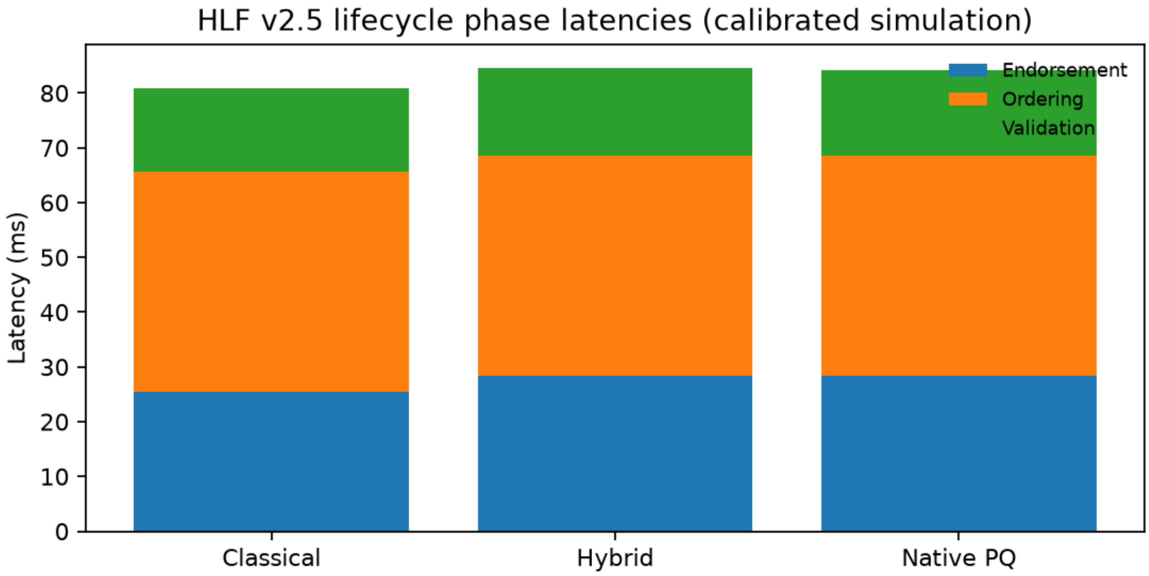

Figure: HLF phase stack (endorsement / ordering / validation).

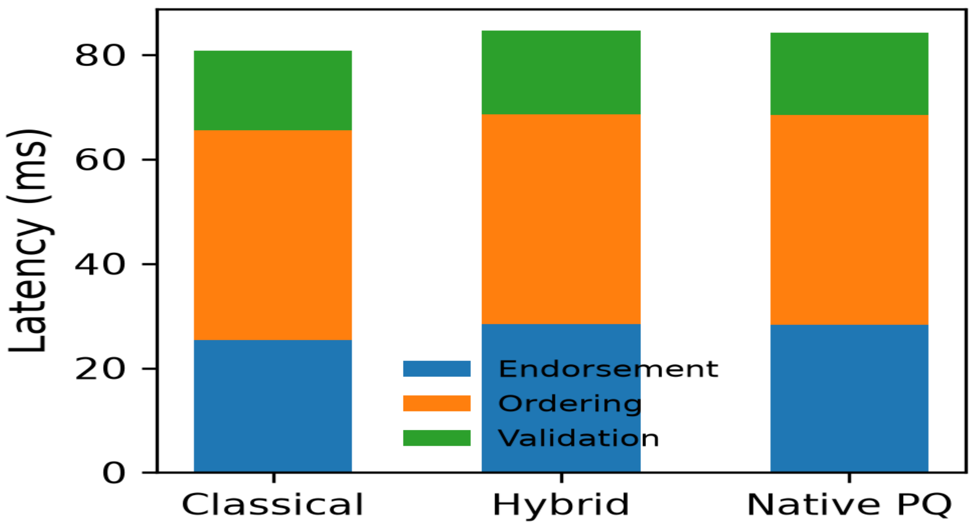

Figure: Pipeline-generated HLF phase chart.

Phase	Classical	Hybrid	Native PQ
Endorsement (ms)	25.4	28.4	28.3
Ordering (ms)	40.2	40.2	40.2
Validation (ms)	15.3	16.0	15.7
Total (ms)	80.8	84.6	84.2

Analysis. Ordering dominates. PQ verification adds a few milliseconds. Live testbed: orderer + Hospital A/B + Research peers + CouchDB Up. Fabric MSP remains ECDSA; ML-DSA applies at application/chaincode payload layer.

9. Forecast Validation & Threat Timeline

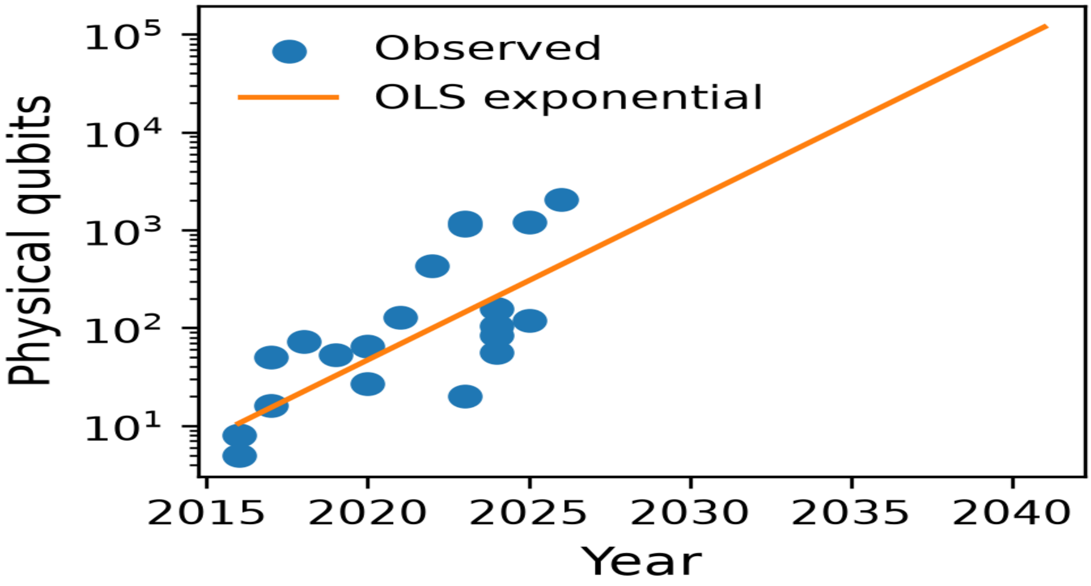

Figure: Physical-qubit growth OLS exponential fit.

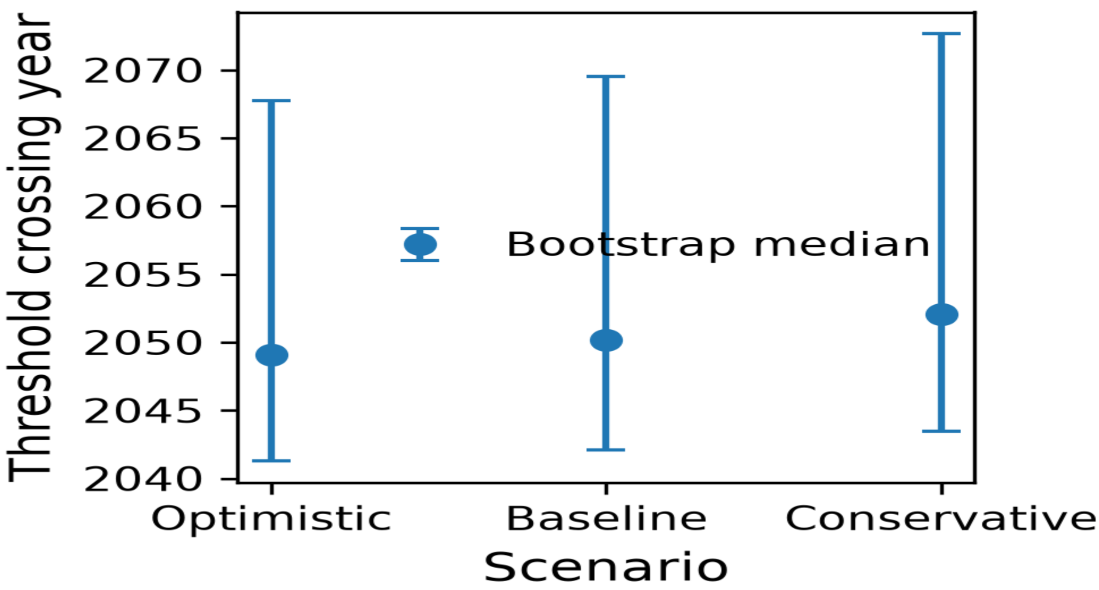

Figure: Bootstrap crossing-year uncertainty by η scenario.

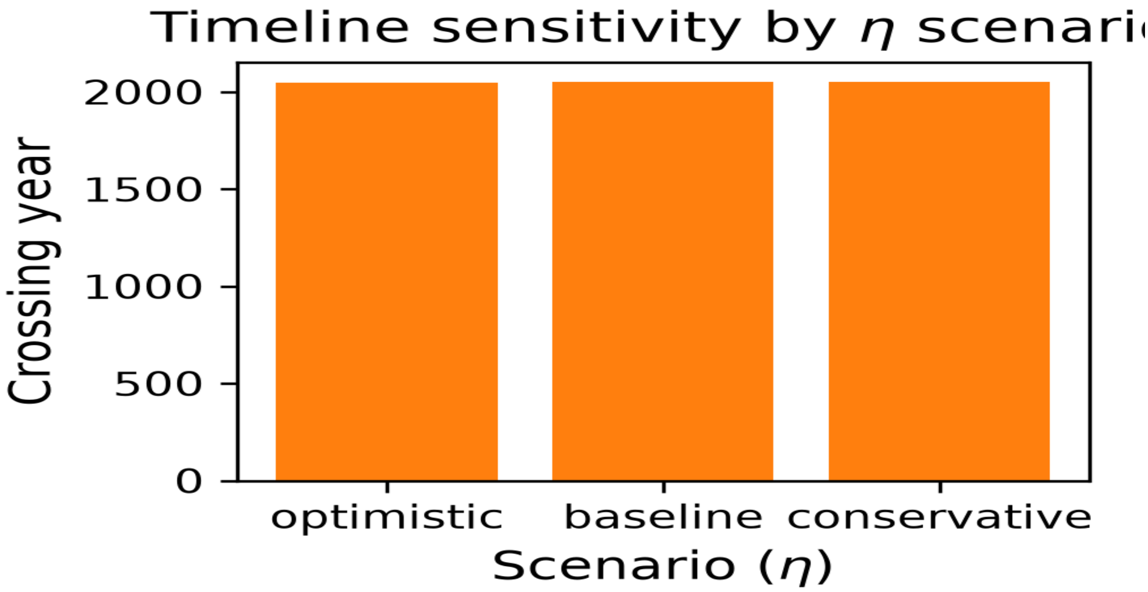

Figure: Timeline sensitivity by η scenario.

Strategy	MAE	RMSE	MAPE (%)
Hold-out 2024–2026	563.39	602.62	424.63
Rolling-origin	600.51	768.92	311.40
Leave-one-out	316.24	544.69	151.37

Analysis. High MAPE reflects sparse, discontinuous hardware jumps—use as planning heuristics. Residual diagnostics support log-linear modeling (Shapiro p=0.395, DW=1.80). Architecture adopts bootstrap 95% lower bound ~2042.1 as the risk-management trigger.

10. Statistical Outcomes

Metric	Comparison	Test	Result
accuracy	all mode pairs	TOST	equivalent
f1	all mode pairs	TOST	equivalent
latency	classical↔hybrid	paired t	not significant
latency	classical↔native	paired t	not significant
latency	hybrid↔native	Wilcoxon	not significant

IID baseline α=1.0, 25 clients, FedAvg, no attack; Holm-corrected α=0.05. Supports utility-neutral and practically latency-neutral PQ modes under this masking design.

11. Cross-Layer Performance Synthesis

Question	Evidence	Outcome
When migrate?	L1 bootstrap ~2042	Start hybrid now (long EHR retention)
Which params?	L2 ablation	ML-KEM-768 / ML-DSA-65 default
FL utility hit?	TOST equivalent	No predictive degradation
FL latency hit?	Holm NS latency	Training dominates crypto tax
Ledger hit?	+4% TX latency	Ordering dominates endorsements
Poisoning?	Exp D	FedAvg fragile; robust aggs need tuning

12. Outputs Checklist

Artifact	Path	State
Macros	results/results_macros.tex	FLNumSeeds=10, FLNumRounds=50
FL tables	results/fl/fl_*.tex	Present
Stats	results/statistics/statistical_tests.tex	Present
Forecast val	results/forecasting/validation/	Present
PQC / ablation	results/pqc/*.tex	Present
Resource	results/resource/resource_profiling.tex	Present
Ledger	results/blockchain/ledger_latency.tex	Present
Figures	figures/output/*.pdf|png	Present
This PDF	docs/QRFL_Complete_Report.pdf	Generated

13. Limitations

- FL PQ cost is modeled from Exp A means, not a live PQ-TLS stack every round.
- Fabric MSP remains ECDSA; PQ authenticity is application/chaincode-level.
- Forecast MAPE is high — prefer lower-bound planning over a single year.
- RAPL energy and Go chaincode deploy remain deferred on this host.
- Byzantine Median/Krum need further tuning for PneumoniaMNIST imbalance.

Sources: qrfl-artifacts/results/*, figures/output/*, ARCHITECTURE_PROCESS_ANALYSIS.md, PHASE_RESULTS_REPORT.md.
Generated automatically by scripts/build_complete_report_pdf.py.